Sources of Magnetic Fields

Building Models to Describe our World

Biot-Savart Law, Ampère’s Law and Solenoids

Outline

Biot-Savart Law

Building Models to Describe our World
Biot-Savart Law, Ampère's Law and Solenoids

Review: Calculating the electric field

Three ways to determine the electric field:

1. Model a charge distribution as the sum of point charges, dq:

$$E = \int dE = \int k\frac{dq}{r^2}$$

2. Exploit symmetry and use Gauss' Law:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

3. Build from common charge distributions (e.g. model a plane as many lines of charge):

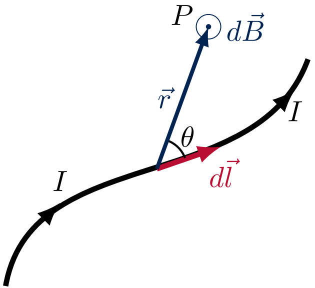
$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

The Biot-Savart Law

A small section of wire, $d\vec{l}$, carrying current, I, will create a magnetic field, $d\vec{B}$, given by:

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \hat{r}}{r^2}$$

- $d\vec{l}$: the small segment (along wire, in direction of I).
- $\vec{r}$: a vector **from the segment** to where we want to calculate the field.
- r: the magnitude of $\vec{r}$.
- $\hat{r}$: unit vector in direction of $\vec{r}$.



Using the Biot-Savart law to determine the magnetic field at point P due to the wire carrying current, I.

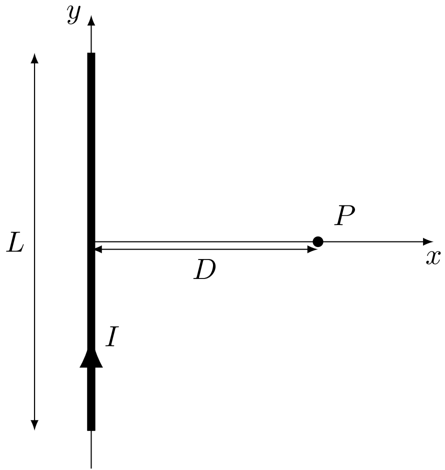
The unit vector

- In practice, it is not always convenient to first calculate a vector, $\vec{r}$, and then a unit vector in that direction. Can use $\vec{r}$ directly in the Biot-Savart law:

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \hat{r}}{r^2} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \frac{\vec{r}}{r}}{r^2} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3}$$

Magnetic field from a straight wire of length L

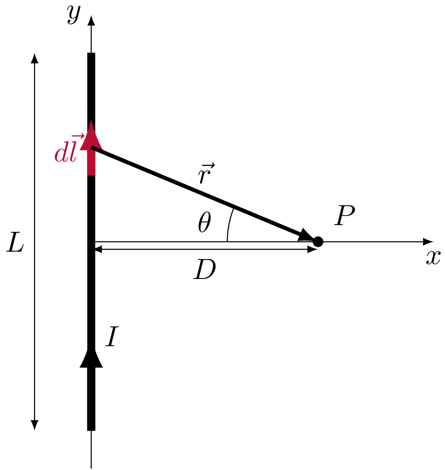
1. Make a good diagram.



Using the Biot-Savart law to determine the magnetic field at point P due to the wire carrying current, I.

Magnetic field from a straight wire of length L

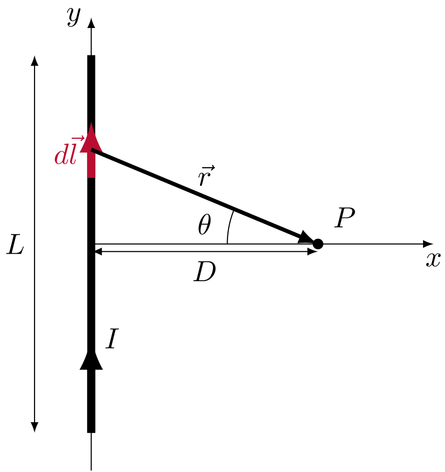
1. Make a good diagram.
2. Choose $d\vec{l}$.



Using the Biot-Savart law to determine the magnetic field at point P due to the wire carrying current, I.

Magnetic field from a straight wire of length L

1. Make a good diagram.
2. Choose $d\vec{l}$.
3. Determine $d\vec{l} \times \vec{r}$:



Using the Biot-Savart law to determine the magnetic field at point P due to the wire carrying current, I.

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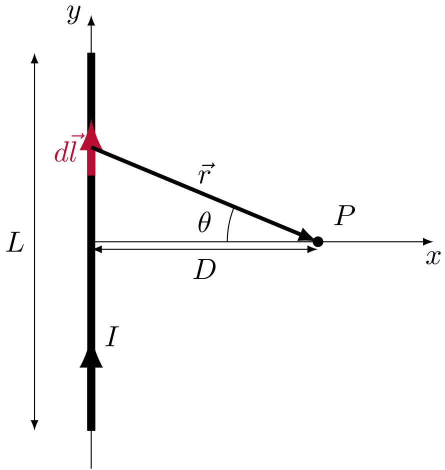
1. Make a good diagram.
2. Choose $d\vec{l}$.
3. Determine $d\vec{l} \times \vec{r}$:

$$d\vec{l} \times \vec{r} = (dl\hat{y}) \times (r\cos\theta\hat{x} - r\sin\theta\hat{y})$$



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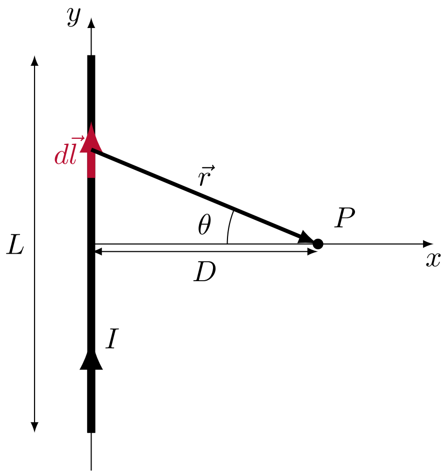

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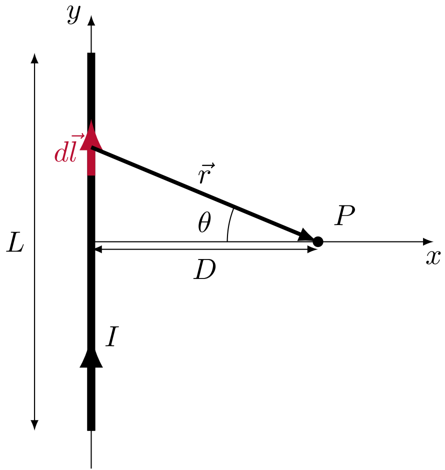

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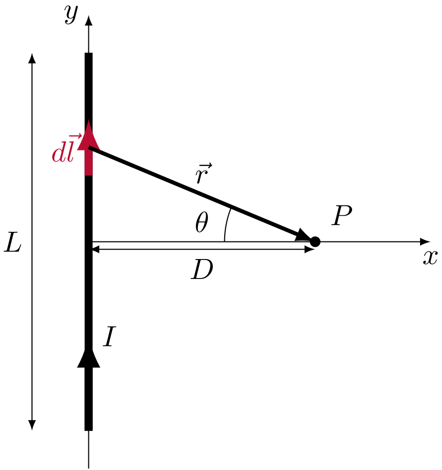

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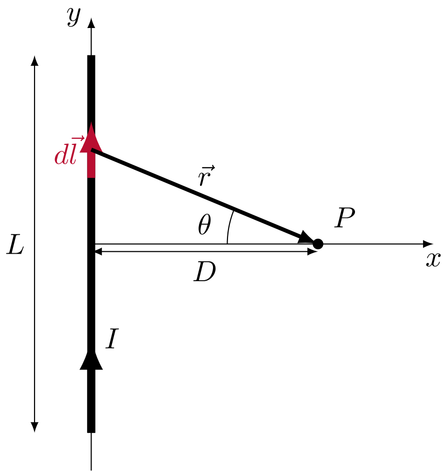

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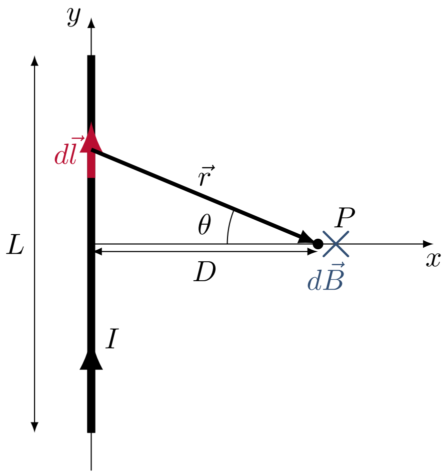
$$d\vec{B} = \frac{\mu_0 I}{4\pi}\frac{d\vec{l} \times \vec{r}}{r^3} = -\frac{\mu_0 I}{4\pi}\frac{\cos\theta}{r^2}dl\hat{z}$$



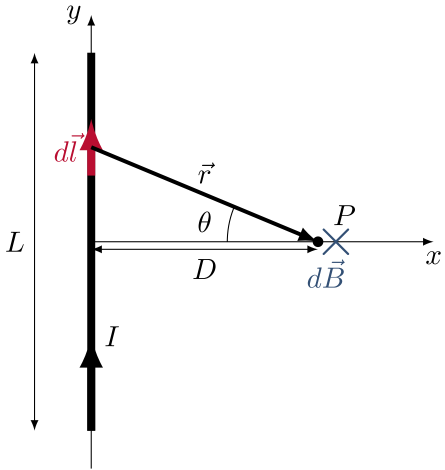
Using the Biot-Savart law to determine the magnetic field at point P due to the wire carrying current, I.

Magnetic field from a straight wire of length L

4. Determine $d\vec{B}$:

$$d\vec{B} = -\frac{\mu_0 I}{4\pi}\cos\theta \frac{1}{r^2} dl\hat{z}$$

5. Need to choose variable of integration (dy, $d\theta$) to integrate.



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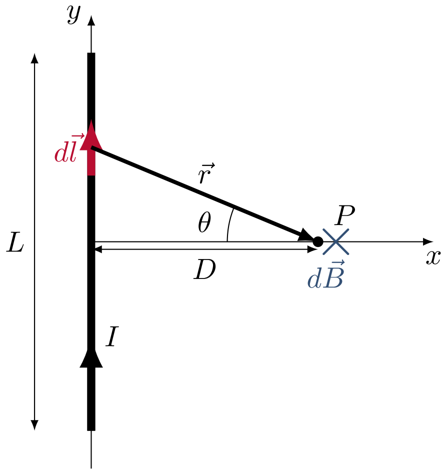

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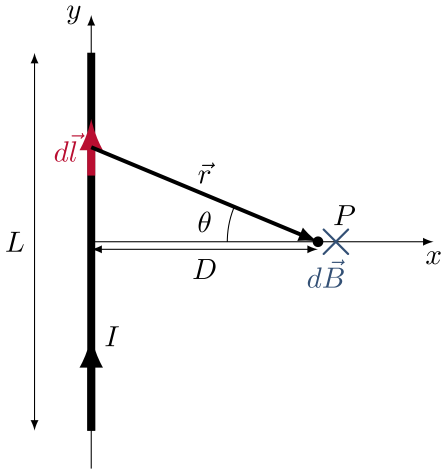

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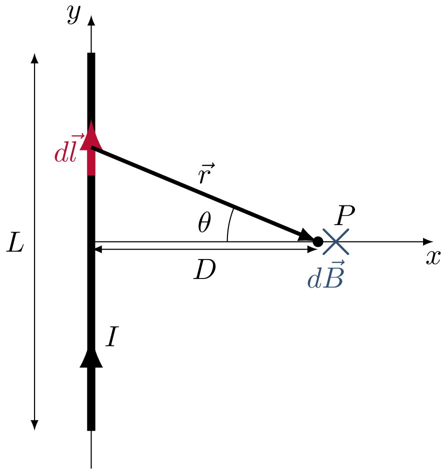
   - $\rightarrow$ Choose $d\theta$:

$$r = \frac{D}{\cos\theta} \quad \rightarrow \quad \therefore \quad \frac{1}{r^2} = \frac{\cos^2}{D^2}$$

$$l = D\tan\theta \quad \rightarrow \quad \therefore \quad dl = \frac{D}{\cos^2\theta} d\theta$$



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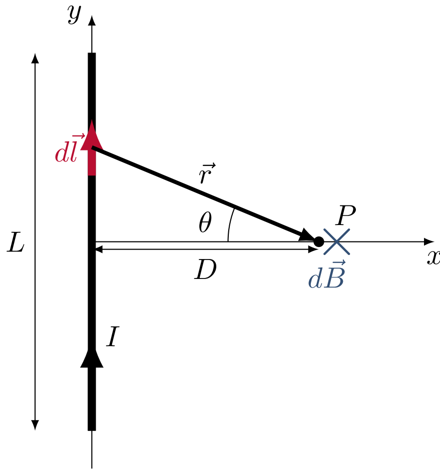
$$\therefore d\vec{B} = -\frac{\mu_0 I}{4\pi}\cos\theta\frac{\cos^2\theta}{D^2}\frac{D}{\cos^2\theta}d\theta\hat{z} = -\frac{\mu_0 I}{4\pi}\frac{\cos\theta}{D}d\theta\hat{z}$$

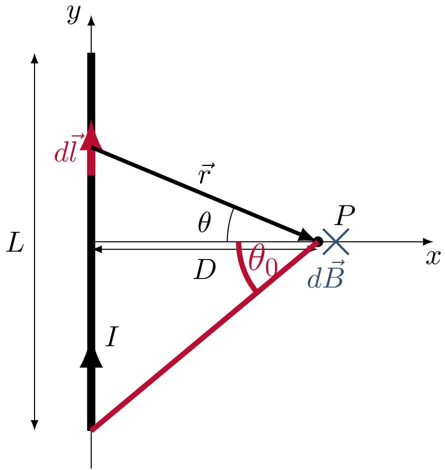


Using the Biot-Savart law to determine the magnetic field at point P due to the wire carrying current, I.

Magnetic field from a straight wire of length L

6. Do the integral:

$$\vec{B} = \int d\vec{B} = -\frac{\mu_0 I}{4\pi D}\int_{-\theta_0}^{+\theta_0} \cos\theta d\theta \hat{z}$$



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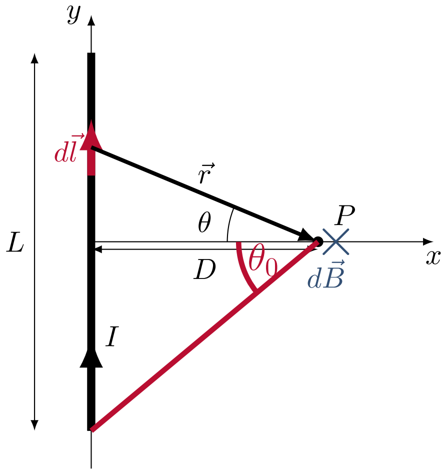
$$= -\frac{\mu_0 I}{4\pi D}\left[\sin\theta\right]_{-\theta_0}^{\theta_0} \hat{z} = -\frac{\mu_0 I}{2\pi D}\sin\theta_0 \hat{z}$$



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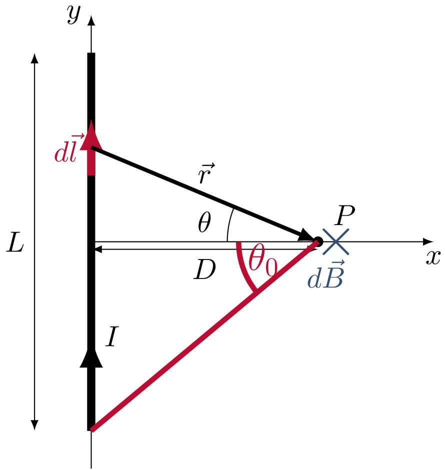
$$\therefore \vec{B} = -\frac{\mu_0 I}{2\pi D}\frac{L/2}{\sqrt{(L/2)^2 + D^2}}\hat{z}$$



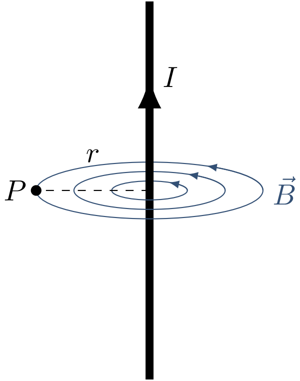
Using the Biot-Savart law to determine the magnetic field at point P due to the wire carrying current, I.

Magnetic field from an infinitely-long wire

Right hand rule for axial vectors, used with thumb parallel to current.

- The field lines are concentric with the wire (right hand rule for axial vectors).
- The magnitude of the field at point, P, a distance, r, from the wire is:

$$B = \frac{\mu_0 I}{2\pi r}$$



Magnetic field forms concentric circles around the wire.

Force between 2 wires

- A wire carrying current produces a magnetic field:

$$B = \frac{\mu_0 I}{2\pi r}$$

- A wire carrying current in a magnetic field feels a force:

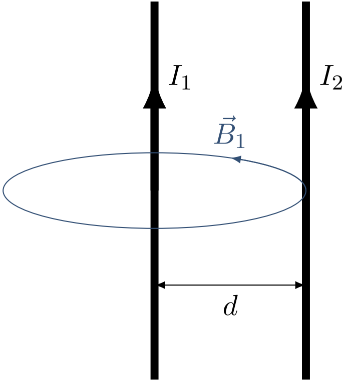
$$\vec{F} = I\vec{l} \times \vec{B}$$

$\rightarrow$ Two wires next to each other both carrying current must exert forces on each other!

Force between two wires

- The field from wire 1:

$$B_1 = \frac{\mu_0 I_1}{2\pi d}$$



Force between two wires

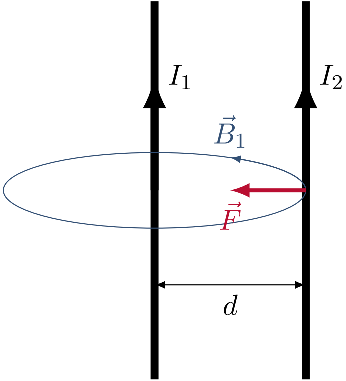
Force between two wires

- The field from wire 1:

$$B_1 = \frac{\mu_0 I_1}{2\pi d}$$

- The force on wire 2:

$$F_2 = I_2 L B_1 = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$



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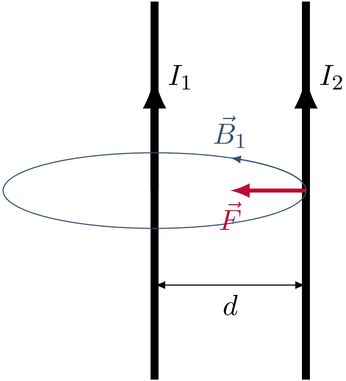
- Is the force on wire 1 from wire 2 the same?



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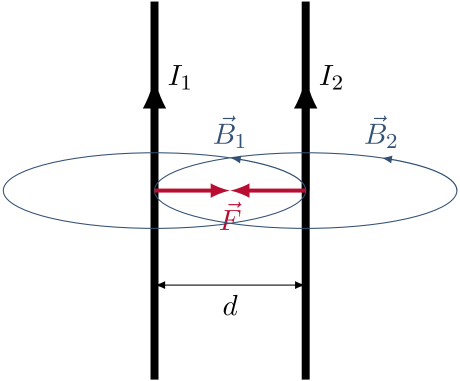
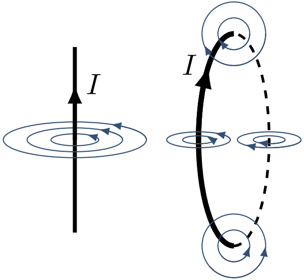
$\rightarrow$ Newton's Third Law...



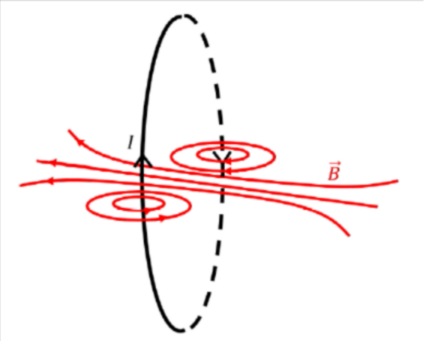
Force between two wires

Field from a loop

- Think of each section of the wire as a small straight wire.
- The field looks like that from a bar magnet.



From the magnetic field of a straight wire to that of a circular loop.



Magnetic field from a ring of current is the same as that of a dipole

.

Magnetic field from a ring of current

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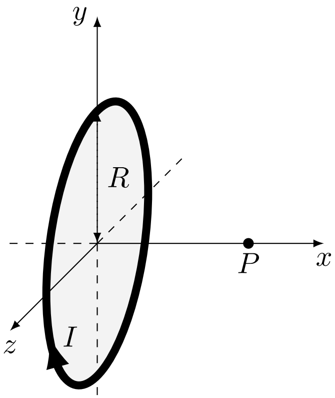
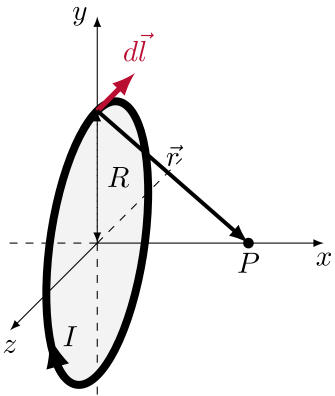


Using the Biot-Savart law to determine the magnetic field at point P due to the ring carrying current, I.

Magnetic field from a ring of current

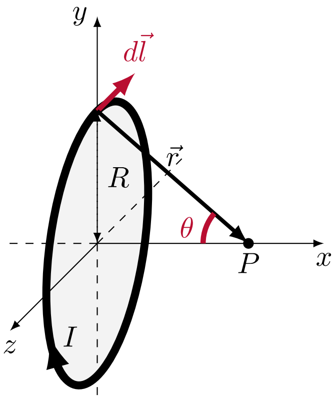
1. Make a good diagram.
2. Choose $d\vec{l}$.



Using the Biot-Savart law to determine the magnetic field at point P due to the ring carrying current, I.

Magnetic field from a ring of current

1. Make a good diagram.
2. Choose $d\vec{l}$.
3. Determine $d\vec{l} \times \vec{r}$:



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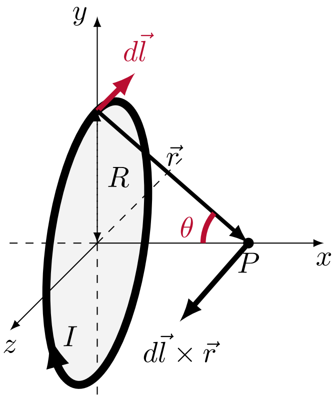
Magnetic field from a ring of current

1. Make a good diagram.
2. Choose $d\vec{l}$.
3. Determine $d\vec{l} \times \vec{r}$:

$$\begin{aligned} d\vec{l} \times \vec{r} &= (-dl\hat{z}) \times (r\cos\theta\hat{x} - r\sin\theta\hat{y}) \\ &= -dlr\cos\theta(\hat{z}\times\hat{x}) + dlr\sin\theta(\hat{z}\times\hat{y}) \\ &= -dlr\cos\theta(\hat{y}) - dlr\sin\theta(\hat{x}) \end{aligned}$$



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Magnetic field from a ring of current

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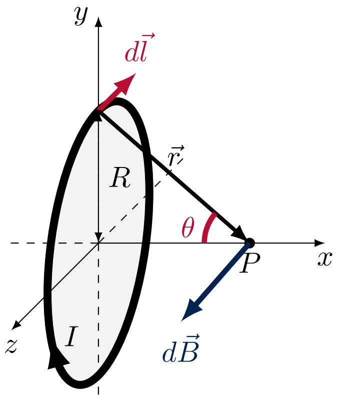
2. Choose $d\vec{l}$.
3. Determine $d\vec{l} \times \vec{r}$: $\rightarrow$ magnitude $= dlr$.
4. Determine $d\vec{B}$:

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3} = -\frac{\mu_0 I dl}{4\pi r^2} (\cos\theta(\hat{y}) + \sin\theta(\hat{x}))$$



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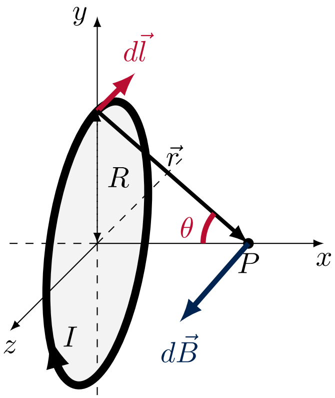

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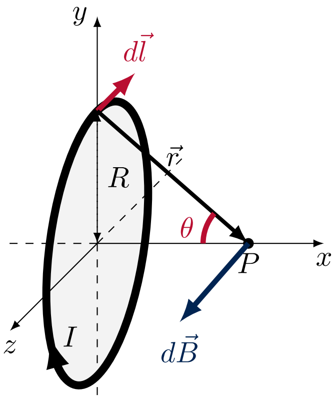

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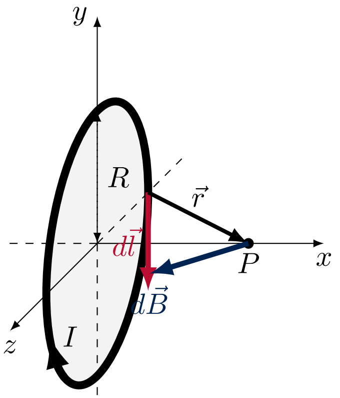

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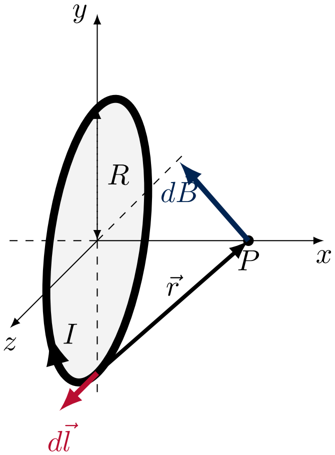

Using the Biot-Savart law to determine the magnetic field at point P due to the ring carrying current, I.

Magnetic field from a ring of current

1. Make a good diagram.
2. Choose $d\vec{l}$.
3. Determine $d\vec{l} \times \vec{r}$: $\rightarrow$ magnitude $= dlr$.
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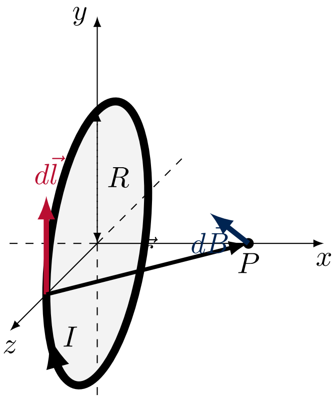

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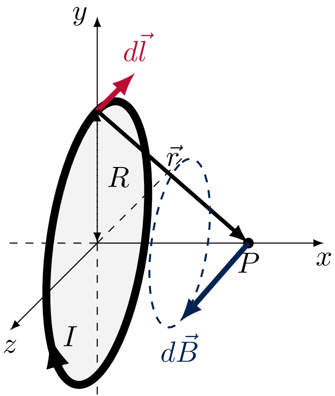
5. Think about symmetry:

The tip of the magnetic field vector traces out a circle. Only the component along $-x$ will survive the integral.



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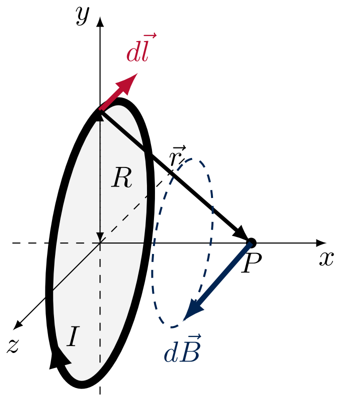

y
$d\vec{l}$
$\vec{r}$
R
x
P
z
I
$d\vec{B}$

Magnetic field from a ring of current

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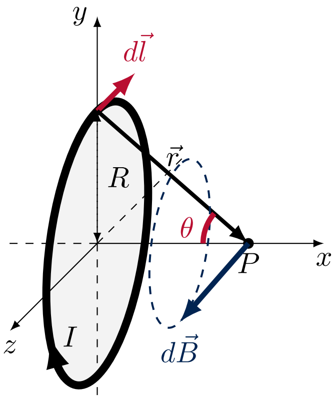
5. Think about symmetry:

Only the x component of the $d\vec{B}$ don't cancel out:

$$dB_x = dB \cos \ldots$$



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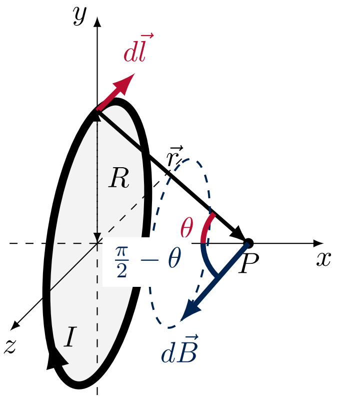
Only the x component of the $d\vec{B}$ don't cancel out:

$$dB_x = dB\cos\left(\frac{\pi}{2} - \theta\right) = dB\sin\theta$$



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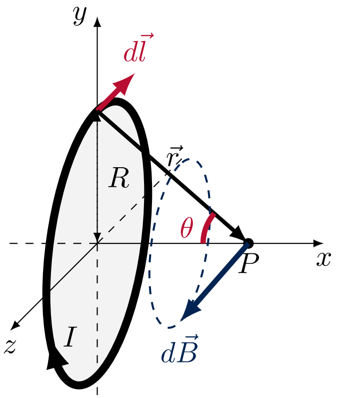
All together:

$$B_x = \int dB_x = \int dB\sin\theta = \frac{\mu_0 I}{4\pi}\int \frac{1}{r^2}\sin\theta dl$$



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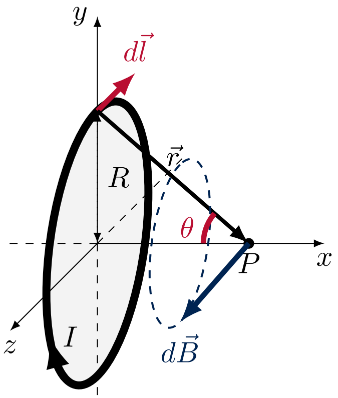

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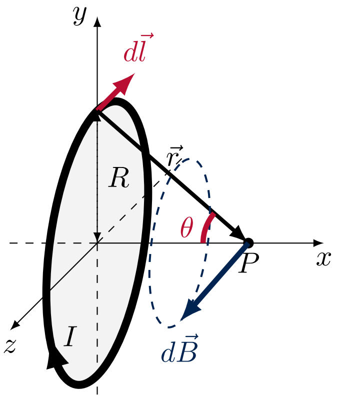
All together:

$$B_x = \int dB_x = \int dB \sin\theta = \frac{\mu_0 I}{4\pi} \int \frac{1}{r^2} \sin\theta dl$$

$$= \frac{\mu_0 I}{4\pi r^2} \sin\theta \int_0^{2\pi R} dl = \frac{\mu_0 I}{4\pi r^2} \sin\theta (2\pi R)$$



Using the Biot-Savart law to determine the magnetic field at point P due to the ring carrying current, I.

Magnetic field from a ring of current

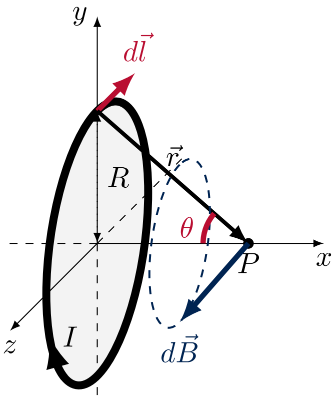
And, finally, with P located a distance x from the origin:

$$B_x = -\frac{\mu_0 I}{4\pi r^2}\sin\theta(2\pi R)$$

$$r = \sqrt{R^2 + x^2}$$

$$\sin\theta = \frac{R}{r} = \frac{R}{\sqrt{R^2+x^2}}$$

$$\boxed{\vec{B} = -\frac{\mu_0 I}{2}\frac{R^2}{(R^2+x^2)^{\frac{3}{2}}}\hat{x}}$$



Using the Biot-Savart law to determine the magnetic field at point P due to the ring carrying current, I.

Ampère’s Law

Building Models to Describe our World
Biot-Savart Law, Ampère’s Law and Solenoids

Ampère's Law

Ampère's Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I^{enc}$$

Similar to Gauss' Law:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q^{enc}}{\epsilon_0}$$

Except:

- Integral is over a path ($d\vec{l}$), not a surface.
- Related to current enclosed by a *closed loop* instead of charge enclosed by a closed surface.

The path integral, $\oint \vec{B} \cdot d\vec{l}$

- We have seen path integrals in the context of work:

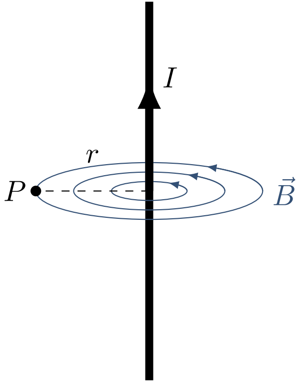
$$W = \int_A^B \vec{F} \cdot d\vec{l}$$

- The circle sign just means that it's a closed path, $A = B$. We know that for a conservative force, $\vec{F}$:

$$\oint \vec{F} \cdot d\vec{l} = 0 \quad \text{(conservative force)}$$

Example: Field near an infinite current-carrying wire

1. Use symmetry to determine the direction of $\vec{B}$.

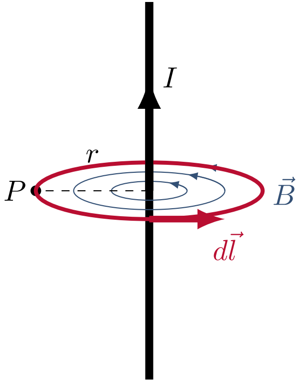


Magnetic field forms concentric circles around the wire.

Example: Field near an infinite current-carrying wire

1. Use symmetry to determine the direction of $\vec{B}$.
2. Choose a path where the integral $\oint \vec{B} \cdot d\vec{l}$ is easy
   - $\vec{B}$ and $d\vec{l}$ make constant angle along path.
   - $\vec{B}$ doesn't change magnitude along path.

   $\rightarrow$ choose a circle of radius r centred on the wire!



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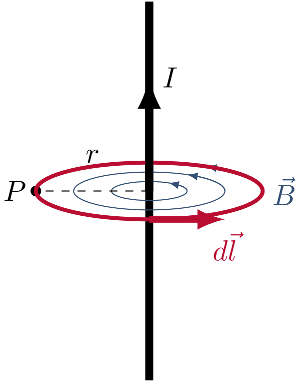
Example: Field near an infinite current-carrying wire

1. Use symmetry to determine the direction of $\vec{B}$.
2. Choose a path where the integral $\oint \vec{B} \cdot d\vec{l}$ is easy
3. Evaluate the integral:

$$\oint \vec{B} \cdot d\vec{l} = \oint B dl = B \oint dl = B 2\pi r$$



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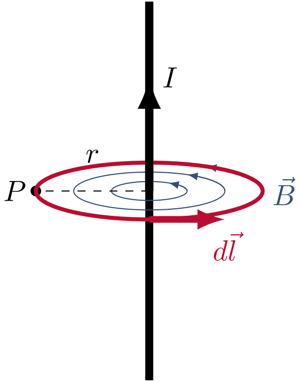
4. Determine the enclosed current: $I^{enc} = I$ (it's all enclosed!)



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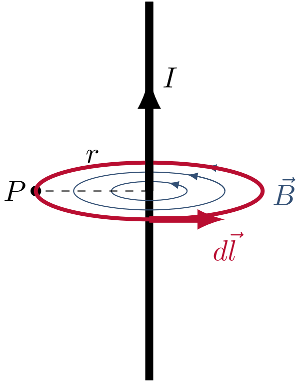
4. Determine the enclosed current: $I^{enc} = I$ (it's all enclosed!)
5. Apply Ampère's Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I^{enc} \ \rightarrow B 2\pi r = \mu_0 I \ \rightarrow B = \frac{\mu_0 I}{2\pi r}$$



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Comparing Gauss and Ampère's Laws

Gauss

- Need high degree of symmetry
- Choose a **surface** that exploits that symmetry:
  - Want field constant in magnitude along surface, and makes a constant angle with surface element, $d\vec{A}$.
- Calculate **flux:**

$$\oint \vec{E} \cdot d\vec{A} = E \cos\theta \oint dA$$

- Calculate **charge** enclosed by the closed surface

Flux represents field lines coming out of a surface.

Ampère

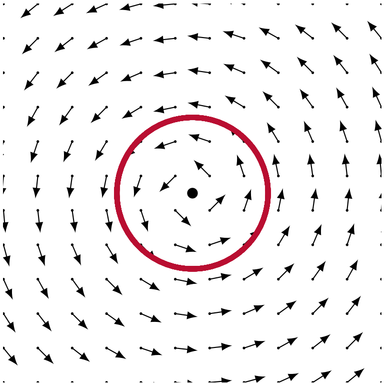
- Need high degree of symmetry
- Choose a **loop** that exploits that symmetry:
  - Want field constant in magnitude along loop, and makes a constant angle with path element, $d\vec{l}$.
- Calculate **circulation:**

$$\oint \vec{B} \cdot d\vec{l} = B \cos\theta \oint dl$$

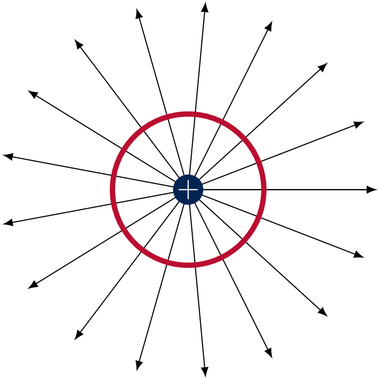
- Calculate **current** enclosed by the closed loop.

Circulation represents rotation in a field.

Circulation of a field



Field with circulation and no flux.



Field with flux and no circulation.

- Circulation picks out the component of the field tangent to the path (scalar product picks out parallel components of vectors)

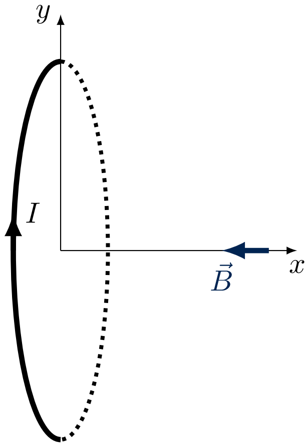
Solenoids

Building Models to Describe our World
Biot-Savart Law, Ampère’s Law and Solenoids

Solenoids

- Field from one loop of current, a distance x from centre along axis of symmetry:

$$\vec{B} = -\frac{\mu_0 I}{2} \frac{R^2}{(R^2 + x^2)^{\frac{3}{2}}} \hat{x}$$



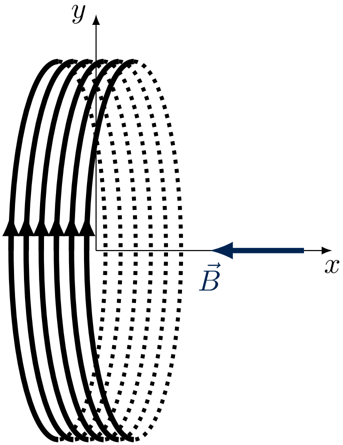
Magnetic field from a circular loop of current.

Solenoids

- Field from one loop of current, a distance x from centre along axis of symmetry:

$$\vec{B} = -\frac{\mu_0 I}{2} \frac{R^2}{(R^2 + x^2)^{\frac{3}{2}}} \hat{x}$$

- By adding more coils, we can get a stronger magnetic field



A solenoid is just a bunch of loops.

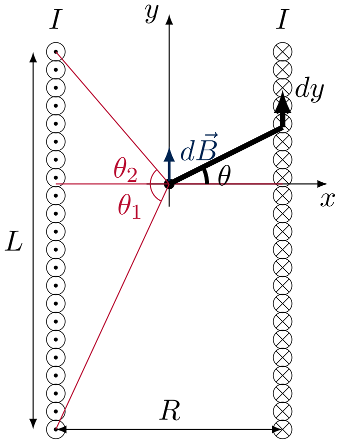
Field inside of a solenoid

- One can calculate the (small) field, $d\vec{B}$, from one circular loop, and then sum the field from each circular loop together:

$$d\vec{B} = \frac{\mu_0 dI}{2} \frac{R^2}{(R^2 + y^2)^{\frac{3}{2}}} \hat{y}$$

$$dI = \left(\frac{NI}{L}\right) dy$$

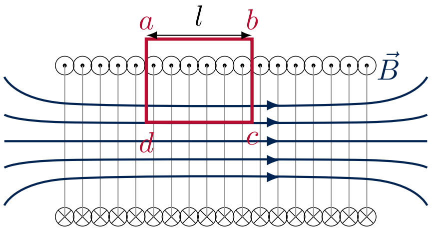
- When you take the limit of the solenoid being very long (compared to its radius), you find that the magnetic field is uniform everywhere inside. Practice!



Field inside of a solenoid

Field inside of a solenoid using Ampère's Law

- The field outside the solenoid is almost zero



Field inside of a solenoid

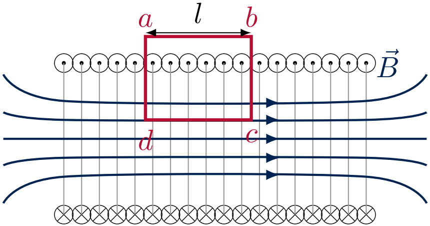
Field inside of a solenoid using Ampère's Law

- The field outside the solenoid is almost zero
- Using Ampère's Law, we have:

$$\oint \vec{B} \cdot d\vec{l} = Bl$$

$$I^{enc} = NI$$

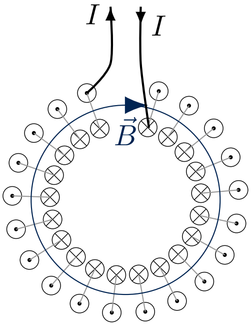
$$\therefore B = \frac{\mu_0 N I}{l} = \mu_0 n I \qquad \left(n = \frac{N}{L} \right)$$



Field inside of a solenoid

The toroid - a solenoid turned into a donut

- This creates a magnetic field with circular field lines.



A toroid.